PROJECT_REPORT

VulNweb: Real-Time Cyber Threat Surveillance System

Project Introduction Report

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Project Type: Project (Supervised Machine Learning + Cybersecurity + Full-Stack Engineering)

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1. Executive Summary

VulNweb is an end-to-end intelligent cybersecurity platform designed to detect malicious or suspicious web activity in real time. The system combines supervised machine learning, explainable AI, heuristic threat scoring, and browser-level integration through a Chrome extension.

The project objective is to provide practical, low-latency protection for users during web navigation by classifying URLs and network activity into actionable threat levels (safe, suspicious, critical), while exposing interpretable explanations for each decision.

At its current stage, the project includes:
- A trained XGBoost threat classification model with strong offline performance
- A FastAPI backend exposing prediction and threat-intelligence endpoints
- A Chrome extension for real-time user-facing alerts and dashboard analytics
- Support tooling for testing, retraining, and deployment

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2. Problem Context and Motivation

Modern users interact with increasingly sophisticated phishing pages, malware delivery URLs, redirects, and social-engineering vectors. Traditional static blacklists are not sufficient for fast-changing threats. This project addresses the gap by building a system that can:
- Learn malicious URL and traffic patterns from labeled data

- Perform quick inference in real-time contexts
- Combine model-driven predictions with explicit security heuristics
- Present understandable threat rationale to improve trust and usability

The final product is designed for practical deployment as a browser-assisted cyber defense mechanism.

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3. Project Objectives

Primary Objectives

- Build a supervised ML model for threat classification
- Achieve high classification quality (target: 95%+ accuracy)
- Integrate model serving via production-style API
- Deliver real-time browser extension monitoring and threat visualization

Secondary Objectives

- Add explainability (SHAP) to improve model interpretability
- Integrate external threat intelligence (VirusTotal)
- Provide feedback and testing loops for iterative improvement
- Prepare deployment artifacts (Docker, Compose, Kubernetes scaffolding)

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4. Data and Feature Engineering

Main Data Sources

- Malicious URLs Dataset (Kaggle): ~651,191 URL samples, 3-class labeling strategy

Data Characteristics

- URL-centric attributes (lengths, tokens, structure, entropy, protocol/domain patterns)
- Multi-class objective:
  - Safe
  - Threat Level 1 (suspicious)

    - Threat Level 2 (highly malicious)

Feature Pipeline

- Threat-oriented URL indicators
- Selected features include lexical, structural, and suspicious-pattern indicators

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5. Modeling Approach

Candidate Models Compared

- Logistic Regression
- Random Forest
- XGBoost (selected)

Reported Evaluation Metrics (Phase 2)

- Logistic Regression: F1 = 0.8283, Accuracy = 83.93%
- Random Forest: F1 = 0.9441, Accuracy = 94.35%
- XGBoost: F1 = 0.9564, Accuracy = 95.69% (best)

Why XGBoost Was Selected

- Best weighted F1 among tested models
- Strong overall accuracy with robust class-wise behavior
- Effective balance between predictive power and deployment practicality

Explainability

- SHAP integration provides top decision drivers per prediction
- Improves transparency for user-facing and debugging scenarios

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6. System Architecture

High-Level Architecture

Chrome Extension → FastAPI Backend → ML Inference Layer → Optional Threat Intelligence / Storage

Main Components

- **Frontend:** Chrome extension (popup, background worker, content script, dashboard, options)
- **Backend:** FastAPI routers for prediction, health, features, model metadata, feedback, and threat endpoints
- **ML Layer:** Serialized XGBoost model with explainability artifacts
- **Threat Intelligence:** VirusTotal API integration for URL/hash analysis

Core API Flows

- Single prediction endpoint for URL analysis
- Batch prediction endpoint for multiple URLs
- Raw-feature endpoint for extension-generated vectors
- Health/model info/features endpoints for observability and integration

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7. Chrome Extension and User Experience

The extension is a key applied-security deliverable and includes:
- Real-time threat indicators (safe/suspicious/critical)
- Threat score and confidence display
- Human-readable explanation snippets
- Desktop notifications for critical events
- Statistics dashboard (distribution, recent checks, counters)
- Local cache and batching logic for performance

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8. Hybrid Threat Scoring Strategy

Beyond pure model output, the project documents a hybrid scoring mechanism combining:
- ML prediction weighting
- Heuristic anomaly indicators (e.g., exfiltration patterns, scanning behavior, long-lived connections, port anomalies, protocol mismatches)

This strategy improves practical resilience by addressing both known learned patterns and potentially novel suspicious behaviors.

Threat level bands are standardized on a 0-100 scale:
- 0-29: Safe

- 30-69: Suspicious
- 70-100: Critical

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9. Performance and Results Summary

Confirmed Performance Signals in Project Documentation

- Best model offline accuracy: 95.69%
- Best model weighted F1: 0.9564
- API response times documented in completion reports as typically under 100ms per request
- System SLA target remains under 500ms for user-facing interaction

Validation and Testing Assets

- Accuracy testing scripts and guide
- Performance testing scripts and guide
- API and integration tests in Python/JS
- Extension behavior and functionality reports

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10. Innovation and Academic Value

This project has high project relevance because it combines:
- Supervised learning for cybersecurity classification
- Explainable AI for trust and auditability
- Real-time systems engineering under latency constraints
- Browser-extension security integration
- API productization and deployment-oriented design

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11. Risks, Limitations, and Mitigation

Observed Risks

- Drift between feature extractor versions and deployed model artifacts
- Potential false positives from aggressive heuristic rules
- Dependency on external threat-intelligence API limits/rate constraints

Mitigations

- Controlled retraining pipeline and metadata versioning
- Confidence thresholds and explanation-driven feedback loops
- Structured test suites (accuracy, performance, extension integration)

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12. Future Work

- Continuous retraining with expanded malicious URL corpora
- Drift detection and automatic model health monitoring
- Improved class-2 (highly malicious) recall optimization
- Broader browser support and enterprise policy controls
- Production observability dashboards (Prometheus/Grafana)
- CI/CD hardening and release automation

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13. Conclusion

VulNweb is a strong project candidate that successfully bridges machine learning theory and practical cybersecurity deployment. The system already demonstrates strong model-level performance, production-style API design, and real-time user integration through a browser extension.

The project is credible, measurable, and extensible, with clear evidence of engineering depth and iterative improvement. It is well-positioned for final presentation, demo defense, and continued research/deployment development.
